Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	04/09/2026
Team ID	6ab4a4a60fc666a751b434a8
Project Name	Auto Ticket Classification using Flow Designer
Maximum Marks	4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

Reference: https://developer.ibm.com/patterns/ai-powered-backend-system-for-order-processing-during-pandemics/

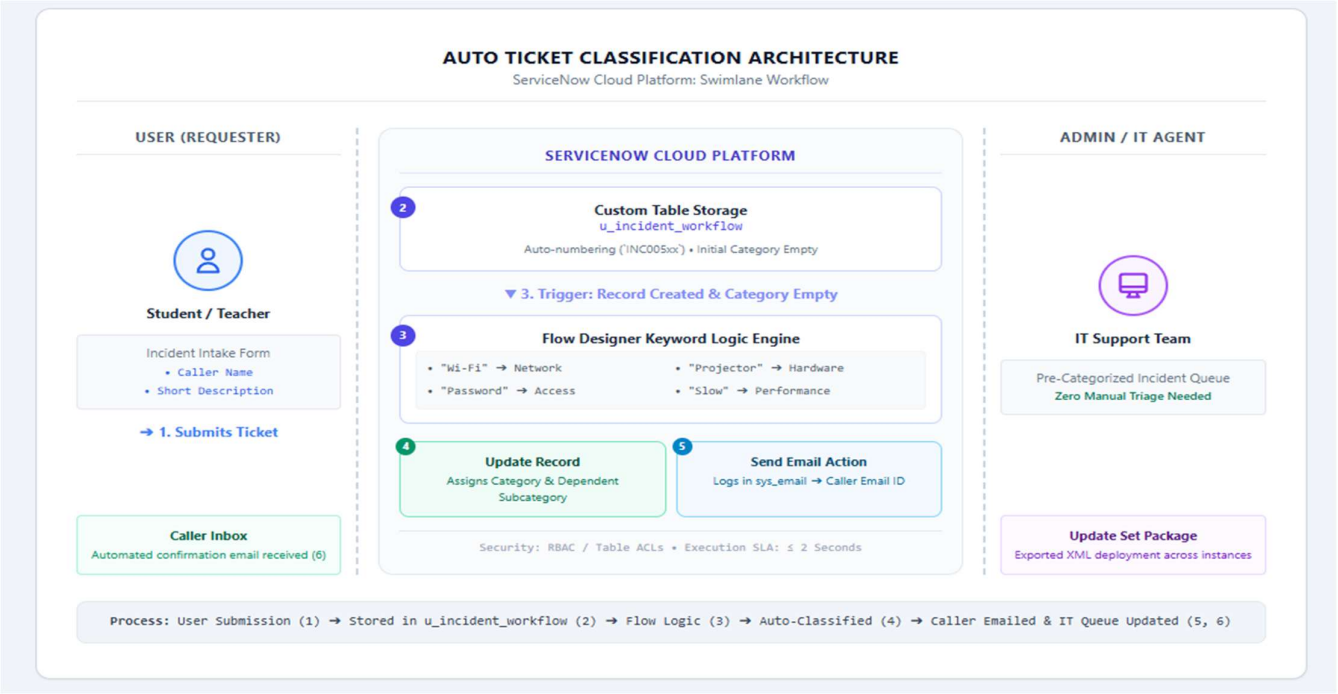


Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	Self-service web form for incident submission with minimal mandatory inputs.	ServiceNow Form Designer / Next Experience UI
2.	Application Logic-1	Flow trigger monitoring record creation when Category is empty.	ServiceNow Flow Designer Trigger Engine
3.	Application Logic-2	Conditional branching (If / Else If) parsing issue keywords in short description.	ServiceNow Flow Logic Engine (No-Code)
4.	Application Logic-3	Dictionary dependency dynamically mapping Subcategories to parent Categories.	ServiceNow Data Dictionary & Dependent Choices
5.	Database	Data Type, Configurations etc.	MySQL, NoSQL, etc.
6.	Notification Service	Automated acknowledgment email generation using dynamic data pills.	ServiceNow Flow Designer Send Email Action
7.	Database Table	Custom incident entity storing auto-numbered records (INC005xx).	Custom Relational Table (u_incident_workflow)
8.	Cloud Database	Enterprise relational database hosting application metadata and schema.	ServiceNow Enterprise Cloud MariaDB Engine
9.	Audit / Email Log	Centralized audit trail capturing outbound notification queue and delivery status.	ServiceNow System Logs (sys_email / SMTP Queue)
10.	Deployment Packaging	Change management container capturing tables, flows, and dictionary entries.	ServiceNow Local Update Set (Project Update Set) / XML
11.	Infrastructure (Cloud)	Enterprise Software-as-a-Service (SaaS) hosting all workflow services.	ServiceNow Cloud Infrastructure (PaaS/SaaS)

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Platform Framework	Standardized cloud framework utilizing no-code flow logic and platform dictionary rules.	ServiceNow Polaris Platform / Glide Engine
2.	Security Implementations	Role-Based Access Control (RBAC) restricting admin configurations and ACL protection on custom tables.	ServiceNow Access Control Lists (ACL) / TLS 1.3 / HTTPS
3.	Scalable Architecture	Asynchronous execution of flow events without blocking frontend intake transactions.	ServiceNow Multi-instance Asynchronous Event Queue
4.	Availability	Cloud-native multi-datacenter deployment ensuring continuous 24/7 service accessibility.	99.9% ServiceNow Cloud Infrastructure SLA
5.	Performance	Automated rule parsing and record updates complete within ≤ 2 seconds of ticket creation.	In-Memory Glide Cache & Indexed Primary Keys

References:

https://c4model.com/

https://developer.ibm.com/patterns/online-order-processing-system-during-pandemic/

https://www.ibm.com/cloud/architecture

https://aws.amazon.com/architecture

https://medium.com/the-internal-startup/how-to-draw-useful-technical-architecture-diagrams-2d20c9fda90d